Constrained Optimality and Duality

Optimization Methods, Week 6

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The constrained problem

Definition (Primal problem)

$$p^* = \inf_{x \in \mathbb{R}^n} f(x) \quad \text{subject to} \quad g_i(x) \leq 0 \ (i = 1, \ldots, m), \quad h_j(x) = 0 \ (j = 1, \ldots, r).$$

The feasible set C contains precisely the points satisfying every constraint.

A local optimum compares $f(x^*)$ only with nearby points in C. Therefore $\nabla f(x^*)$ need not vanish at a boundary optimum.

Active constraints

Definition (Active set)

At a feasible x, $A(x) = \{i : g_i(x) = 0\}$. An inequality with $g_i(x) < 0$ is inactive.

For the disk $g(x) = x_1^2 + x_2^2 - 1 \leq 0$, every point on the circle has an active constraint; every point in the interior has none. The active set describes the local boundary. It does not tell us whether the corresponding multiplier is strictly positive.

The Lagrangian

Definition (Lagrangian)

$$\mathcal{L}(x, \lambda, \nu) = f(x) + \sum_{i=1}^{m} \lambda_i g_i(x) + \sum_{j=1}^{r} \nu_j h_j(x), \qquad \lambda \geq 0, \quad \nu \in \mathbb{R}^r.$$

The multipliers combine objective and constraint gradients. At a feasible point,

$$\mathcal{L}(x, \lambda, \nu) \leq f(x),$$

because $\lambda_i g_i(x) \leq 0$. This simple inequality drives weak duality.

KKT conditions

Theorem (KKT necessity)

At a local minimum where a suitable constraint qualification holds, multipliers (λ^*, ν^*) *satisfy*

$$
\begin{aligned}
\nabla_x \mathcal{L}(x^*, \lambda^*, \nu^*) &= 0 && && \textit{stationarity},\\
g_i(x^*) &\le 0, \quad && h_j(x^*) = 0 && \textit{primal feasibility},\\
\lambda_i^* &\ge 0 && && \textit{dual feasibility},\\
\lambda_i^* g_i(x^*) &= 0 && && \textit{complementary slackness}.
\end{aligned}
$$

Examples of qualifications are LICQ locally and Slater's condition for convex problems.

Complementary slackness in plain terms

$$\lambda_i g_i(x^*) = 0 \quad \Longleftrightarrow \quad \begin{cases} g_i(x^*) < 0 \;\Rightarrow\; \lambda_i = 0, \\ \lambda_i > 0 \;\Rightarrow\; g_i(x^*) = 0. \end{cases}$$

The reverse of the second implication fails: an active constraint can have multiplier zero. For $\min_x x^2$ subject to $-x \leq 0$, the optimum $x^* = 0$ is active and its KKT multiplier is 0.

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A boundary example

Minimize $f(x) = \frac{1}{2}(x-2)^2$ subject to $x - b \le 0$, with $b < 2$.

$$\mathcal{L}(x, \lambda) = \tfrac{1}{2}(x-2)^2 + \lambda(x-b).$$

Stationarity gives $x - 2 + \lambda = 0$. The active candidate is

$$x^* = b, \qquad \lambda^* = 2 - b > 0.$$

Convexity makes this the global optimum. The constraint blocks movement toward the unconstrained minimizer $x = 2$.

The dual function

Definition (Dual function and dual problem)

$$q(\lambda, \nu) = \inf_x \mathcal{L}(x, \lambda, \nu), \qquad d^* = \sup_{\lambda \geq 0,\, \nu \in \mathbb{R}^r} q(\lambda, \nu).$$

For each fixed x, $\mathcal{L}(x, \lambda, \nu)$ is affine in the multipliers. Its pointwise infimum q is concave, even if the primal problem is nonconvex. The dual may return $-\infty$ for some multiplier choices.

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The duality gap

For feasible x and $\lambda \geq 0$,

$$0 \leq f(x) - q(\lambda, \nu).$$

This gap is a certificate: when it is small, the feasible x is nearly optimal. Weak duality holds without convexity or constraint qualifications. A strictly positive optimal gap is possible in nonconvex problems.

Strong duality and Slater's condition

Theorem (Convex strong duality)

Suppose f, g_i are finite-valued convex functions on $\mathbb{R}^n$, $h(x) = Ax - b$ with A of full row rank, the primal optimum is finite, and there exists $\bar{x}$ with $g_i(\bar{x}) < 0$ for all inequality constraints and $h(\bar{x}) = 0$. Then $p^* = d^*$ and the dual optimum is attained.

Strict feasibility is called Slater's condition. It ensures that the multiplier description supports the optimum rather than merely bounding it from below.

A complete primal-dual calculation

For $f(x) = \frac{1}{2}(x-2)^2$ and $g(x) = x - b$, minimize the Lagrangian over x:

$$\partial_x \mathcal{L} = x - 2 + \lambda = 0 \quad \Rightarrow \quad x = 2 - \lambda.$$

Substitution gives

$$q(\lambda) = \lambda(2-b) - \tfrac{1}{2}\lambda^2, \qquad \lambda \geq 0.$$

If $b < 2$, maximizing yields $\lambda^* = 2 - b$ and $d^* = \frac{1}{2}(2-b)^2 = p^*$.

Multiplier as sensitivity

Let $p(b)$ be the optimal value when the bound is $x \leq b$. For $b < 2$,

$$p(b) = \tfrac{1}{2}(b-2)^2, \qquad p'(b) = b - 2 = -\lambda^*.$$

Increasing b relaxes the constraint and decreases the minimum. The multiplier measures the marginal benefit of that relaxation. More generally, for a right-hand side $g_i(x) \leq b_i$,

$$\frac{\partial p}{\partial b_i} = -\lambda_i^*$$

when the value function is differentiable and the usual regularity conditions hold.

Sensitivity when the active set changes

In the same example, $p(b) = 0$ for $b \geq 2$. At $b = 2$, $\lambda^* = 0$ and the first derivative remains continuous, but the second derivative changes. In more general problems, a change in the active set can create a kink in $p(b)$. Then a single derivative need not exist; optimal multipliers describe supporting slopes or one-sided marginal values.

Equality example: Euclidean projection

Minimize $(x-2)^2 + (y-1)^2$ subject to $x + y = 1$.

$$\mathcal{L} = (x-2)^2 + (y-1)^2 + \nu(x + y - 1).$$

Stationarity gives $2(x-2) + \nu = 0$ and $2(y-1) + \nu = 0$. Together with $x + y = 1$,

$$(x^*, y^*) = (1, 0), \qquad \nu^* = 2.$$

The objective gradient $(-2, -2)$ is normal to the feasible line. The equality multiplier has no sign restriction.

A two-dimensional inequality example

Minimize $(x-2)^2 + (y-1)^2$ subject to $x + y \leq 1.5$.

$$\mathcal{L} = (x-2)^2 + (y-1)^2 + \lambda(x + y - 1.5).$$

The active KKT equations give

$$x = 2 - \lambda/2, \quad y = 1 - \lambda/2, \quad x + y = 1.5.$$

Thus $(x^*, y^*) = (1.25, 0.25)$ and $\lambda^* = 1.5$. Direct substitution verifies feasibility, multiplier sign, stationarity, and complementary slackness.